

1. $F = 1.09 \text{ N} \therefore F = Eq, \Delta V = Ed \therefore$ to $1 e^-$ the gun supplies $E = 70. \text{ keV}$, its $\Delta V = 70. \text{ kV}$.
(Chapter 19, Section (2) Electric Potential in a Uniform Electric Field - cp1e_ch19_ex5)
2. $Q = 3.89 \times 10^{-9} \text{ C} \therefore C = \frac{Q}{V} = \epsilon_0 \frac{A}{d}$
(Chapter 19, Section (5) Capacitors and Dielectrics - cp1e_ch19_ex8)
3. $\Delta V_{\max} = 3.90 \times 10^9 \text{ V} \therefore \Delta V = Ed$
(Chapter 19, Section (2) Electric Potential in a Uniform Electric Field - cp1e_ch19_ex4)
4. $\varepsilon = 14.6 \text{ V} \therefore \varepsilon = NAB\omega \sin(\omega t), \sin(\omega t) = 1$
(Chapter 23, Section (5) Electric Generators - cp1e_ch23_ex4)
5. $\beta = 95.1 \text{ dB} \therefore v_w = \left(331 \frac{\text{m}}{\text{s}}\right) \sqrt{\frac{T}{273 \text{ K}}}, I = \frac{(\Delta p)^2}{2\rho v_w}, \beta(\text{dB}) = 10 \log_{10}\left(\frac{I}{I_0}\right), I_0 = 1 \times 10^{-12} \frac{\text{W}}{\text{m}^2}$
(Chapter 17, Section (3) Sound Intensity and Sound Level - cp1e_ch17_ex2)
6. $D = 7.31 \times 10^{-5} \text{ m} \therefore R = \frac{\rho L}{A}$
(Chapter 20, Section (3) Resistance and Resistivity - cp1e_ch20_ex5)
7. $\lambda = 2.31 \times 10^{-6} \text{ m} \therefore d \sin \theta = m\lambda$
(Chapter 27, Section (3) Young's Double-Slit Experiment - cp1e_ch27_ex1)
8. $L = 0.4733 \text{ m} \therefore f_n = n \frac{v_w}{4L}, v_w = \left(331 \frac{\text{m}}{\text{s}}\right) \sqrt{\frac{T}{273 \text{ K}}}$
(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5a)
9. $m = -0.387 \therefore \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
(Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex6b)
10. $x_3 = 0.866 \text{ m} \therefore \Sigma \vec{F}_x = m \vec{a}_x, F = k \frac{|q_1 q_2|}{r^2}$
(Chapter 18, Section (3) Coulomb's Law - PSE_Serway_4e_ch23_ex3)
11. $\theta_c = 38.7^\circ \therefore n_1 \sin \theta_1 = n_2 \sin \theta_2$
(Chapter 25, Section (4) Total Internal Reflection - cp1e_ch25_ex4)
12. $f = 4.37 \times 10^6 \text{ Hz} \therefore T = \frac{1}{f}$
(Chapter 16, Section (2) Period and Frequency in Oscillations - cp1e_ch16_ex3)
13. $I_p = 3.57 \text{ A} \therefore \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$
(Chapter 23, Section (7) Transformers - cp1e_ch23_ex6)

$$14. f = 4.65 \times 10^8 \text{ Hz} \therefore \lambda_{obs} = \lambda_s \sqrt{\frac{1 + \frac{u}{c}}{1 - \frac{u}{c}}}, c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{E}{B} = f\lambda$$

(Chapter 28, Section (4) Relativistic Addition of Velocities - cp1e_ch28_ex5b)

$$15. B = 0.209 \text{ T} \therefore r = \frac{mv}{qB}, m_p = 1.67 \times 10^{-27} \text{ kg}, q_p = 1.60 \times 10^{-19} \text{ C}$$

(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - tkd1)

$$16. E_{ind} = 1.08 \text{ J} \therefore E_{ind} = \frac{1}{2} L I^2$$

(Chapter 23, Section (9) Inductance - cp1e_ch23_ex8)

$$17. a = 0.895 \% \therefore Z = \rho v, a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2}$$

(Chapter 17, Section (7) Ultrasound - cp1e_ch17_ex7)

$$18. R = 5.82 \Omega \therefore V = IR$$

(Chapter 20, Section (2) Ohm's Law - cp1e_ch20_ex4)

$$19. P = 7.69 \text{ D} \therefore P = \frac{1}{f}$$

(Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex5)

$$20. B_{max} = 1.30 \times 10^{-6} \text{ T} \therefore c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{E}{B} = f\lambda$$

(Chapter 24, Section (2) Production of Electromagnetic Waves - cp1e_ch24_ex1)

$$21. I_s = 2.61 \times 10^{-3} \text{ A} \therefore \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

(Chapter 23, Section (7) Transformers - cp1e_ch23_ex5)

$$22. L = 3.58 \text{ cm} \therefore T_{pend} = 2\pi \sqrt{\frac{L}{g}}$$

(Chapter 16, Section (4) Simple Pendulum - cp1e_ch16_p22)

$$23. C_T = 2.20 \times 10^{-9} \text{ F} \therefore \frac{1}{C_s} = \frac{1}{C_1} + \dots + \frac{1}{C_n}$$

(Chapter 19, Section (6) Capacitors in Series and Parallel - cp1e_ch19_ex9)

$$24. Q = 1.35 \times 10^3 \text{ C} \therefore I = \frac{\Delta Q}{\Delta t}$$

(Chapter 20, Section (1) Current - tkd0)

$$25. f = 1.12 \times 10^3 \text{ Hz} \therefore f_n = n \frac{v_w}{4L}, v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}$$

(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5b)

26. $F = 1.35 \times 10^{-12} \text{ N} \therefore F = qvb \sin \theta, \theta = 90^\circ$
(Chapter 22, Section (4) Magnetic Field Strength - cp1e_ch22_ex1)
27. $I = 1.18 \times 10^3 \text{ A} \therefore I = \frac{\Delta Q}{\Delta t}$
(Chapter 20, Section (1) Current - cp1e_ch20_ex1)
28. $k = 5.88 \times 10^4 \frac{\text{N}}{\text{m}} \therefore \vec{F} = -k \Delta \vec{x}, k = -\frac{\vec{F}}{\Delta \vec{x}}$
(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex1)
29. $P = 52.6 \text{ D} \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
(Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex2a)
30. $F = 3.89 \times 10^{-6} \text{ N} \therefore F = qvb \sin \theta \therefore F_{\max} = qvb$
(Chapter 22, Section (4) Magnetic Field Strength - cp1e_ch22_p7)
31. $I = 4.96 \text{ A} \therefore \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}, V = IR$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex1)
32. $I_{\text{RMS}} = 75.0 \text{ A} \therefore P_{\text{ave}} = I_{\text{RMS}} V_{\text{RMS}}$
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ch20_ex10)
33. $\varepsilon = 1.41 \times 10^{-4} \text{ V} \therefore \varepsilon = B l v$
(Chapter 22, Section (6) The Hall Effect - cp1e_ch22_p22)
34. $I = 5.98 \times 10^{-12} \text{ A} \therefore I(t) = I_0 e^{-t/\tau}, \tau = \frac{L}{R}$
(Chapter 23, Section (10) RL Circuits - cp1e_ch23_ex9b)
35. $n_2 = 1.16 \therefore n_1 \sin \theta_1 = n_2 \sin \theta_2$
(Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex2)
36. $P = -4.42 \text{ D} \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
(Chapter 26, Section (2) Vision Correction - cp1e_ch26_ex3)
37. $\varepsilon = 9.52 \times 10^3 \text{ V} \therefore \varepsilon = B l v$
(Chapter 23, Section (3) Motional EMF - cp1e_ch23_ex2)
38. $I = 0.543 \text{ A} \therefore$ Kirchhoff's rules
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ1)
39. $E = 3.16 \times 10^3 \text{ J} \therefore P = IV = \frac{E}{t}, V = \text{emf} - Ir$
(Chapter 21, Section (2) Electromotive Force: Terminal Voltage - cp1e_ch21_ex4)
40. $t = 7.75 \times 10^{-8} \text{ m} \therefore 2t = \left(m + \frac{1}{2}\right) \lambda_n, n_{\text{air}} < n_{\text{oil}} > n_{\text{water}}$

(Chapter 27, Section (7) Thin-Film Interference - cp1e_ch27_p72)

41. $P = 1.38 \text{ W} \therefore R_s = R_1 + R_2 + R_3, V = IR, P = IV$

(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex2)

42. $\frac{F}{l} = 140 \cdot \frac{\text{N}}{\text{m}} \therefore F = IlB \sin \theta, \theta = 90^\circ$

(Chapter 22, Section (7) Magnetic Force on a Current-Carrying Conductor - cp1e_ch22_ex4)

43. $\dot{N}_e = 4.19 \times 10^{20}$ electrons per second $\therefore$

$$\Delta V = \frac{\Delta PE}{q}, W_{c.f.} = \Delta PE \therefore W = -q \Delta V \therefore \frac{q}{t} = \frac{-W}{t \Delta V} \therefore \dot{N}_e = \frac{q}{q_e t} = \frac{-W}{q_e t \Delta V}$$

(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cp1e_ch19_ex2)

44. $f = 0.671 \text{ Hz} \therefore T_{SHO} = 2\pi \sqrt{\frac{m}{k}}, f = \frac{1}{T}$

(Chapter 16, Section (3) Simple Harmonic Motion - cp1e_ch16_ex4)

45. $D = 1.40 \times 10^{-6} \text{ m} \therefore D \sin \theta = m\lambda$

(Chapter 27, Section (5) Single-Slit Diffraction - cp1e_ch27_ex4)

46. $P = 52.6 \text{ D} \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$

(Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex2b)

47. $\tau = 0.0157 \text{ s} \therefore \tau = \frac{L}{R}$

(Chapter 23, Section (10) RL Circuits - cp1e_ch23_ex9)

48. $\text{emf} = 95.7 \text{ V} \therefore \text{emf} = -N \frac{\Delta \Phi}{\Delta t}$

(Chapter 23, Section (2) Faraday's Law of Induction: Lenz's Law - cp1e_ch23_p8)

49. $L = 5.34 \times 10^{-5} \text{ H} \therefore L = N \frac{\Delta \Phi}{\Delta t} = \frac{\mu_0 N^2 A}{l}$

(Chapter 23, Section (9) Inductance - cp1e_ch23_ex7)

50. $E = 1.17 \times 10^{14} \text{ J} \therefore E_0 = mc^2$

(Chapter 28, Section (6) Relativistic Energy - cp1e_ch28_ex6)

51. $E = 4.68 \times 10^4 \text{ N/C} \therefore E = k \frac{|Q|}{r^2}$

(Chapter 18, Section (4) Electric Field - cp1e_ch18_ex2)

52. $v_f = 4.78 \times 10^5 \frac{\text{m}}{\text{s}} \therefore \Delta V = \frac{\Delta PE}{q}, W_{c.f.} = -\Delta PE \therefore W = -q \Delta V, W_{net} = \Delta KE \therefore \Delta KE = -q \Delta V$

(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cp1e_ch19_ex3)

53. $m = -1.67 \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$

(Chapter 26, Section (6) Image Formation by Lenses - cp1e_ch25_ex7)

54. $B = 1.05 \times 10^{-4} \text{ T} \therefore r = \frac{mv}{qB}, m_e = 9.11 \times 10^{-31} \text{ kg}, q_e = 1.60 \times 10^{-19} \text{ C}$
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cp1e_ch22_p13)
55. $I = 2.03 \text{ A} \therefore$ Kirchhoff's rules
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ2)
56. $\theta = 0.0411 \text{ arcseconds} \therefore \theta = 1.22 \frac{\lambda}{D}$
(Chapter 27, Section (6) Limits of Resolution - cp1e_ch27_ex5)
57. $\theta_2 = 3.15^\circ \therefore n_1 \sin \theta_1 = n_2 \sin \theta_2$
(Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex3)
58. $\% \Delta P = -23.9\% \therefore \rho(T) = \rho_0(1 + \alpha \Delta T), R = \frac{\rho L}{A}, P = \frac{V^2}{R}$
(Chapter 20, Section (4) Electric Power and Energy - tkd3)
59. $Q = 3.30 \times 10^{-6} \text{ C} \therefore V = \frac{kQ}{r}$
(Chapter 19, Section (3) Electric Potential Due to a Point Charge - cp1e_ch19_ex7)
60. $\Delta V = 17.5 \text{ V} \therefore V = \frac{kQ}{r}$
(Chapter 19, Section (3) Electric Potential Due to a Point Charge - cp1e_ch19_ex6)
61. $v = 2.31 \times 10^8 \frac{\text{m}}{\text{s}} \therefore n = \frac{c}{v}$
(Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex1)
62. $\Delta y = 4.63 \times 10^{-12} \text{ m} \therefore F = qvb \sin \theta, F_{\text{net}} = ma, \vec{x}(t) = \vec{x}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2$
(Chapter 22, Section (11) Applications of Magnetism - cp1e_ch22_p82)
63. $C_T = 1.06 \times 10^{-8} \text{ F} \therefore \frac{1}{C_s} = \frac{1}{C_1} + \dots + \frac{1}{C_n}, C_p = C_1 + \dots + C_n$
(Chapter 19, Section (6) Capacitors in Series and Parallel - cp1e_ch19_ex10)
64. $x = 4.25 \times 10^3 \text{ m} \therefore \theta = 1.22 \frac{\lambda}{D}, \Delta \theta = \frac{\Delta s}{r}, \Delta s \approx 0.70 \text{ m}$ for small angles
(Chapter 27, Section (6) Limits of Resolution - cp1e_ch27_p66)
65. $W_{\text{ratio}} = 171 \cdot \text{times more energy} \therefore \Delta V = \frac{\Delta PE}{q}, W = -\Delta PE$ for a conservative force, so $W = -q \Delta V$
(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cp1e_ch19_ex1)
66. $d = 1.03 \times 10^{-6} \text{ m} \therefore P = \frac{V^2}{R}, R = \frac{\rho L}{A}$ and $A = \pi r^2$
(Chapter 20, Section (4) Electric Power and Energy - tkd2)

$$67. E_0 = 1.67 \times 10^3 \frac{\text{V}}{\text{m}} \therefore I_{ave} = \frac{c \epsilon_0 E_0^2}{2} = \frac{c B_0^2}{2 \mu_0} = \frac{E_0 B_0}{2 \mu_0}$$

(Chapter 24, Section (4) Energy in Electromagnetic Waves - cp1e_ch24_ex4)

$$68. v_{max} = 0.363 \frac{\text{m}}{\text{s}} \therefore v(t) = \sqrt{\frac{k}{m}} X \sin \frac{2\pi t}{T} \text{ and the max of sine's range is 1.}$$

(Chapter 16, Section (5) Energy and the Simple Harmonic Oscillator - cp1e_ch16_ex6)

$$69. \mathcal{E} = 7.35 \text{ V} \therefore \mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

(Chapter 23, Section (5) Electric Generators - cp1e_ch23_ex3)

$$70. d_i = 0.186 \text{ m} \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

(Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex6a)

$$71. h_i = -1.99 \times 10^{-5} \text{ m} \therefore \frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$$

(Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex1)

$$72. q = 2.78 \times 10^{-8} \text{ C} \therefore \Sigma \vec{F}_x = m \vec{a}_x, F = k \frac{|q_1 q_2|}{r^2}$$

(Chapter 18, Section (3) Coulomb's Law - PSE_Serway_4e_ch23_ex5)

$$73. \Delta t = 1.34 \times 10^{-6} \text{ s} \therefore \gamma = \left(\sqrt{1 - \frac{v^2}{c^2}} \right)^{-1}, \Delta t = \gamma \Delta t_0$$

(Chapter 28, Section (2) Simultaneity and Time Dilation - cp1e_ch28_ex1)

$$74. r = 0.491 \text{ m} \therefore r = \frac{mv}{qB}, m_p = 1.67 \times 10^{-27} \text{ kg}, q_p = 1.60 \times 10^{-19} \text{ C}$$

(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cp1e_ch22_ex2)

$$75. \theta = 71.6^\circ \therefore I = I_0 \cos^2 \theta$$

(Chapter 27, Section (8) Polarization - cp1e_ch27_p87)

$$76. F = 2.99 \times 10^{-5} \text{ N} \therefore F = Eq$$

(Chapter 18, Section (4) Electric Field - cp1e_ch18_ex3)

$$77. R_{eq} = 2.12 \Omega \therefore R_s = R_1 + R_2 + R_3, \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}$$

(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex3)

$$78. P = 10.6 \text{ D} \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$$

(Chapter 26, Section (2) Vision Correction - cp1e_ch26_ex4)

$$79. L = 1.82 \text{ ly} \therefore L = L_0 \gamma^{-1}$$

(Chapter 28, Section (3) Length Contraction - cp1e_ch28_ex2)

$$80. f = 560.5 \text{ Hz} \therefore f_{obs} = f_s \left(\frac{v_w}{v_w \pm v_s} \right) \text{ (review the Doppler effect)}$$

(Chapter 17, Section (4) Doppler Effect and Sonic Booms - cp1e_ch17_ex4)

$$81. I = 12.0 \text{ A} \therefore B = \frac{\mu_0 I}{2\pi r}$$

(Chapter 22, Section (9) Magnetic Fields produced by Currents: Ampere's Law - cp1e_ch22_ex6)

$$82. d = 0.787 \text{ m} \therefore \Delta E_t = W_{nc} = -f_k d, f_k = \mu_k N = \mu_k mg, PE = \frac{1}{2} k x^2$$

(Chapter 16, Section (7) Damped Harmonic Motion - cp1e_ch16_ex7)

$$83. \varepsilon = 1.63 \times 10^{-4} \text{ V} \therefore \varepsilon = B l v$$

(Chapter 22, Section (6) The Hall Effect - cp1e_ch22_ex3)

$$84. F = 4.64 \times 10^{-6} \text{ N} \therefore F = k \frac{|q_1 q_2|}{r^2}$$

(Chapter 18, Section (3) Coulomb's Law - cp1e_ch18_ex1)

$$85. F = 25.1 \text{ N} \therefore \frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

(Chapter 22, Section (10) Magnetic Force between Two Parallel Conductors - cp1e_ch22_p50)

$$86. PE = 0.510 \text{ J} \therefore PE = \frac{1}{2} k x^2$$

(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex2)

$$87. E = 3.36 \times 10^7 \frac{\text{N}}{\text{C}} \therefore E = k \frac{|Q|}{r^2}, E^2 = E_x^2 + E_y^2$$

(Chapter 18, Section (5) Electric Field Lines - cp1e_ch18_ex4)

$$88. d_{min} = 2.25 \times 10^{-3} \text{ m} \therefore R = \frac{\rho L}{A}, P_{\text{avg}} = I^2 R$$

(Chapter 20, Section (4) Electric Power and Energy - tkd4)

$$89. I = 21.0 \text{ A} \therefore B = \frac{\mu_0 I}{2R}$$

(Chapter 22, Section (9) Magnetic Fields produced by Currents: Ampere's Law - cp1e_ch22_ex7)

$$90. E = 8.70 \times 10^3 \text{ J} \therefore P = IV = \frac{E}{t}$$

(Chapter 20, Section (4) Electric Power and Energy - tkd1)

$$91. \theta = 24.9^\circ \therefore d \sin \theta = m\lambda$$

(Chapter 27, Section (4) Multiple-Slit Diffraction - cp1e_ch27_ex3)

$$92. d = 823. \text{ m} \therefore \vec{v}_{avg} = \frac{\Delta \vec{x}}{\Delta t}$$

(Chapter 17, Section (2) Speed of Sound, Frequency, Wavelength - cp1e_ch17_p8)

$$93. \lambda_{obs} = 0.696 \text{ m} \therefore \lambda_{obs} = \lambda_s \sqrt{\frac{1 + \frac{u}{c}}{1 - \frac{u}{c}}}$$

(Chapter 28, Section (4) Relativistic Addition of Velocities - cp1e_ch28_ex5)

$$94. C = 1.95 \times 10^{-6} \text{ F} \therefore E_{cap} = \frac{C V^2}{2}$$

(Chapter 19, Section (7) Energy Stored in Capacitors - cp1e_ch19_ex11)

$$95. F = 1.35 \times 10^{-12} \text{ N} \therefore F = Eq$$

(Chapter 18, Section (8) Applications of Electrostatics - cp1e_ch18_ex5)

$$96. P = 57.6 \text{ W} \therefore P = \frac{V^2}{R}$$

(Chapter 20, Section (4) Electric Power and Energy - cp1e_ch20_ex7)

$$97. \tau = 18.9 \text{ N} \cdot \text{m} \therefore \tau = NIAB \sin \theta \therefore \tau_{max} = NIAB$$

(Chapter 22, Section (8) Torque on a Current Loop: Motors and Meters - cp1e_ch22_ex5)

$$98. V_0 = 255. \text{ V} \therefore V_{RMS} = \frac{V_0}{\sqrt{2}}$$

(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ch20_ex9)